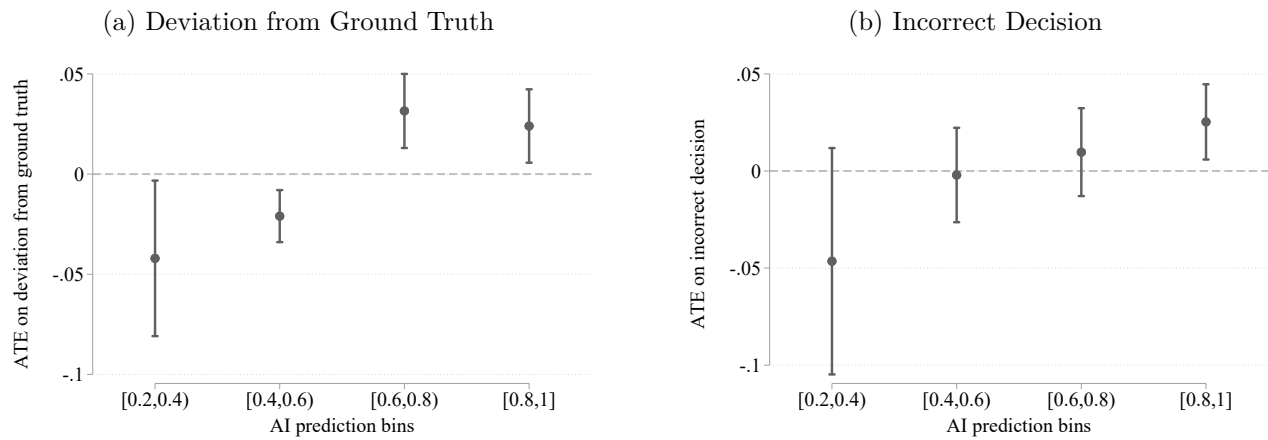


Figure C.37: Conditional treatment effect given AI prediction



Note: Main specifications similar to figure 3 with the exception that only the abnormal pathology is considered.

Table C.17: Average Treatment Effects

	Deviation from AI		Deviation from Ground Truth		
	Pooled with AI	Abnormal	Pooled	Pooled with AI	Abnormal
Treatment	(1)	(2)	(3)	(4)	(5)
AI $\times$ CH	0.000 (0.001)	0.001 (0.005)	0.001 (0.001)	0.001 (0.002)	0.009 (0.008)
AI	-0.016 (0.001)	-0.054 (0.005)	-0.001 (0.001)	0.001 (0.002)	0.009 (0.007)
CH	-0.000 (0.001)	0.001 (0.004)	-0.001 (0.001)	-0.003 (0.002)	-0.010 (0.006)
CONTROL MEAN	0.108 (0.003)	0.271 (0.010)	0.033 (0.002)	0.085 (0.005)	0.421 (0.013)
PATHOLOGY FE	Yes	No	Yes	Yes	No
OBSERVATIONS	235820	18140	1850280	235820	18140

Note: Main specifications similar to table 2 with the exception that only the abnormal pathology is considered.

C.5 Automation Bias Appendix

C.5.1 Conditional Independence

Table C.18 presents evidence that human and AI signals are not conditionally independent. To test the hypothesis of conditional independence, we regress the human report in the treatment arm without AI assistance on the ground truth, the AI score, and the interaction between ground truth and the AI score. If the signals were conditionally independent, we would observe the AI score to offer no predictive power on the human report after conditioning on the ground truth. As shown in Table C.18 we can reject this null hypothesis.

Table C.18: Test of Conditionally Independent Signals

	Top Level with AI	Pooled with AI	Abnormal
GROUND TRUTH	0.318 (0.042)	0.265 (0.033)	-0.049 (0.078)
AI SCORE	0.536 (0.046)	0.620 (0.035)	0.815 (0.052)
GROUND TRUTH $\times$ AI	-0.244 (0.082)	-0.217 (0.066)	0.205 (0.090)
CONSTANT	0.089 (0.011)	0.044 (0.003)	-0.058 (0.039)
OBSERVATIONS	11420	57100	5710
R-SQUARED	0.260	0.301	0.316

Note: Estimates of a regression of the human report in the treatment without AI assistance on the ground truth interacted with the AI score. Standard errors are two-way clustered at the radiologist and patient case level.

C.5.2 Estimating Bayesian Update Terms

Here, we describe the method we use to estimate the Bayesian benchmark $\pi(\omega_i = 1 | s_{ir}^E, s_i^A)$. This procedure is done separately for each pathology. We train a random forest classifier that predicts the ground truth based on features including the vector of a radiologist’s reported probabilities in the non-AI treatment and the vector of AI predictions. Additional features include radiologist identifiers to allow for heterogeneity in radiologists’ assessments, an indicator equal to one if the case was read with clinical history, and summaries of the patient clinical history. We estimate this quantity for various parameterizations of s_{ir}^E and s_i^A described in Section 5. These are used in the model testing exercise to understand if radiologists account for the joint distribution of signals when forming their posterior beliefs. The hyperparameters of the model are tuned using grouped cross-validation where observations were grouped by patient id to avoid overfitting with five folds. We impose monotonicity constraints on the model to impose that $\pi(\omega_i = 1 | s_{ir}^E, s_i^A)$ is monotonically increasing in all probability inputs. When the model includes clinical history, we provide a summarized patient’s clinical record with their sex, weight, temperature, pulse, age, and the number of available and flagged labs. We mean impute these variables when the radiologist does not have access to the clinical history and include an indicator equal to one if the radiologist had access to the clinical history as an additional feature. Below we summarize the performance of these models and the relative value of increasing the dimension of s_{ir}^E and s_i^A .